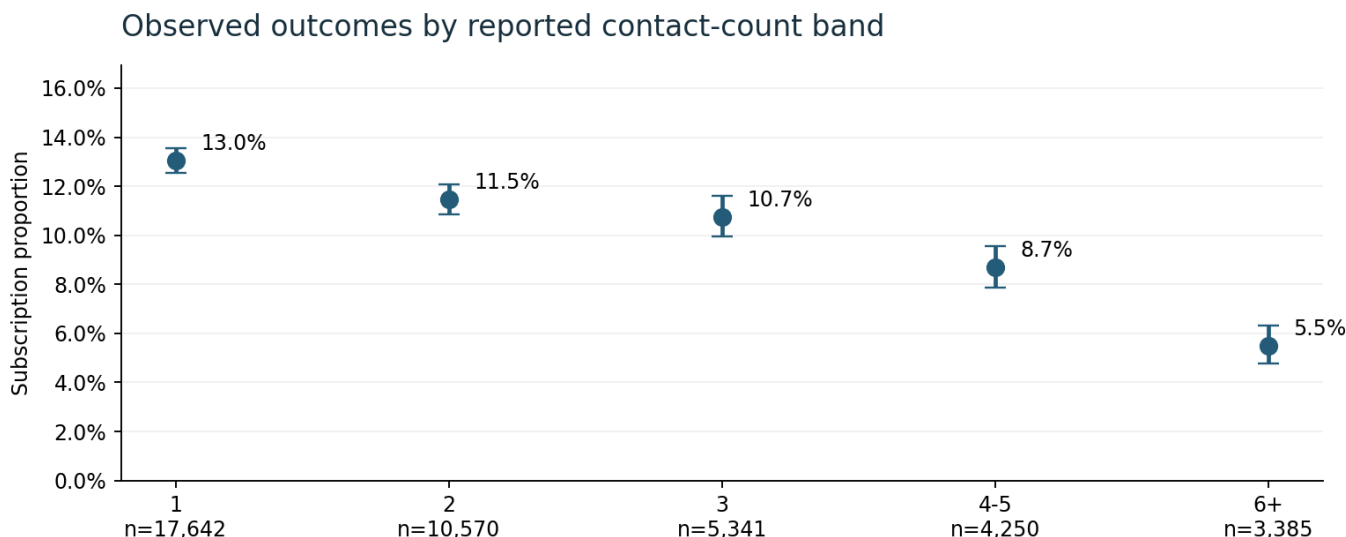


Contact pressure & subscription outcomes

Independent portfolio analysis | Portuguese bank campaigns | Historical source: May 2008-November 2010

41,188 records | 4,640 subscriptions | 11.27% observed proportion



Records with one reported current-campaign contact show a 13.04% subscription proportion; the 6+ contact group shows 5.49%. These are different observed groups, not an experiment.

Decision supported: review repeat-contact rules and investigate selection/stopping behavior. A controlled pilot with identified customers, contact timestamps and agreed outcomes is needed before choosing a contact cap.

Do not infer: that fewer contacts cause higher conversion, or that a cap would deliver a quantified saving. Successful subscriptions may themselves stop further contacts.

Points show record-level proportions; bars show 95% Wilson intervals. Repeated clients cannot be identified. Independence may fail, making these intervals too narrow.

Compare like with like

Channel comparison: association, not treatment effect



Channel rates differ, but channel was not randomly assigned. Prior campaign history and pooled calendar-month composition can differ too. Use the exported contact-by-prior-outcome strata to investigate these associations.

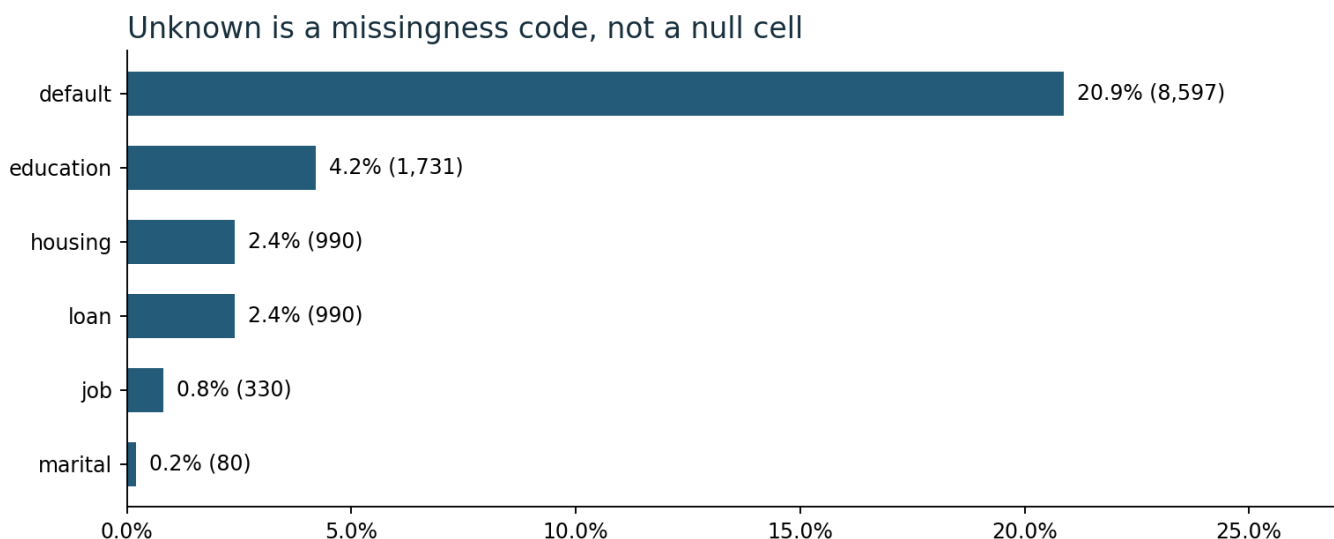
Previous campaign outcome	Records	Subscriptions	Observed rate
failure	4,252	605	14.23%
nonexistent	35,563	3,141	8.83%
success	1,373	894	65.11%

Available evidence: `pressure_strata.csv` retains every contact-band/channel/prior-outcome group, including small samples. `channel_by_month.csv` pools matching calendar months across years; it is not a chronological monthly series.

Next measurement: assign stable customer and campaign IDs, preserve every contact timestamp, record actual costs and define a follow-up window. These additions permit customer-level workload analysis and a defensible pilot; they are recommendations, not implemented outcomes.

The file lacks unique customer identifiers and exact dates. Reported contact counts include the last contact; their sum is not a verified count of distinct calls. No ROI, revenue or staffing benefit is estimated.

Quality before conclusions



12 excess identical rows retained. Without business keys, identical records are not proven ingestion duplicates. Dropping them could remove legitimate observations.

History ambiguity: 4,110 records have pdays=999 but positive previous contacts. The dictionary describes 999 as no previous contact; these records conflict with that interpretation. They remain flagged, not rewritten. Sentinel values are excluded from numeric recency summaries.

Leakage boundary: call duration is known after the contact. It is excluded from targeting and prospective recommendations. This project is descriptive operations analysis, not a predictive model.

Reproduction: run `python build.py`, the unittest suite, then `python render_report.py`. Source fingerprint, controls and query outputs accompany this brief. The executed notebook exposes the main quality checks and SQL results.

Source: Moro, S., Rita, P., & Cortez, P. (2014). Bank Marketing. UCI Machine Learning Repository. DOI: 10.24432/C5K306. CC BY 4.0.

UCI source and license | Dataset variant: bank-additional-full.csv, 20 inputs plus outcome.